Advanced Calculus

Limits of Functions

Two-Sided Limits ・ One-Sided Limits and Limits at Infinity

雙邊極限 ・ 單邊極限與無窮遠處的極限

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§ 3.1

Two-Sided Limits

雙邊極限

Definition: Limit of a Function

Definition 3.1

Let $a \in \mathbb{R}$, let I be an open interval containing a, and let f be defined on I except possibly at a. Then $f(x) \to L$ as $x \to a$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

Two features of the definition

- I **is open**: f is defined on both sides of a (for δ small).
- $0 < |x - a|$: the value $f(a)$ plays no role; f need not even be defined at a.

ε：允許的誤差；δ：x 要離 a 多近。δ 可以依賴 ε、f、a，但**不能依賴** x。

Compare with Sequences

Sequence: $x_n \to L$ vs. Function: $f(x) \to L$

$$x_n \to L: \quad \forall \varepsilon > 0\ \exists N \in \mathbb{N}:\ n \geq N \ \Rightarrow\ |x_n - L| < \varepsilon$$

$$f(x) \to L: \quad \forall \varepsilon > 0\ \exists \delta > 0:\ 0 < |x - a| < \delta \ \Rightarrow\ |f(x) - L| < \varepsilon$$

Negation: $f(x) \not\to L$ as $x \to a$

$$\exists\, \varepsilon_0 > 0\ \ \forall\, \delta > 0\ \ \exists\, x:\ 0 < |x - a| < \delta \ \text{ and } \ |f(x) - L| \geq \varepsilon_0$$

「$n \geq N$」對應到「$0 < |x - a| < \delta$」：都是在描述「夠靠近終點」。否定時取 $\delta = \frac{1}{n}$ 就能造出一個數列。

How to Find δ

課本 Example 3.2、3.3 的手法

Linear case (Example 3.2): $f(x) = mx + b, m \neq 0$

$|f(x) - f(a)| = |m|\,|x - a|$, so take $\delta = \varepsilon/|m|$.

General strategy (Example 3.3: $x^2 + x - 3 \to -1$ as $x \to 1$)

1. **Factor out** $|x - a|$**:** $\quad f(x) - L = x^2 + x - 2 = (x - 1)(x + 2)$.
2. **Restrict** $\delta \leq 1$ and bound the other factor:
   $|x - 1| < 1 \Rightarrow 0 < x < 2 \Rightarrow |x + 2| < 4$.
3. **Combine:** $\quad |f(x) - L| < 4|x - 1|$, so take

$$\delta = \min\left\{1,\ \frac{\varepsilon}{4}\right\}.$$

關鍵：把「比較不重要的因子」用一個常數上界取代；$\min$ 的兩個條件都要用到。

Simplifying Before Taking Limits

Remark 3.4

If $f(x) = g(x)$ for all $x \in I \setminus \{a\}$ and $f(x) \to L$ as $x \to a$, then $g(x) \to L$ as $x \to a$.

Example 3.5

$$g(x) = \frac{x^3 + x^2 - x - 1}{x^2 - 1} = \frac{(x+1)(x^2-1)}{x^2-1} = x + 1 \quad (x \neq \pm 1)$$

so $\lim_{x \to 1} g(x) = \lim_{x \to 1} (x + 1) = 2$.

- Algebra that is invalid at **finitely many points** (e.g. cancelling $x - a$) is allowed.
- The value of the limit is independent of $f(a)$.

因為定義只看 $x \neq a$，所以可以先約分、化簡，再求極限。

Sequential Characterization of Limits

Theorem 3.6

Let $a \in \mathbb{R}$, I an open interval containing a, and f defined on I except possibly at a. Then

$$\lim_{x \to a} f(x) = L \iff f(x_n) \to L \text{ for } \textbf{every} \text{ sequence } x_n \in I \setminus \{a\} \text{ with } x_n \to a.$$

Two uses

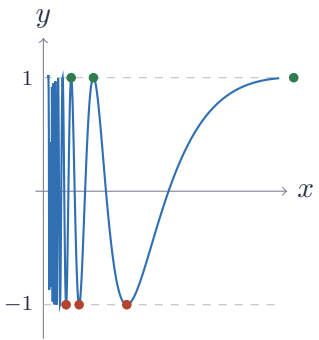
- **Transfer theorems:** results for sequences (Ch. 2) $\Rightarrow$ results for functions.
- **Show a limit does NOT exist:** find two sequences $a_n, b_n \to a$ ($a_n, b_n \neq a$) with

  $$\lim_{n \to \infty} f(a_n) \neq \lim_{n \to \infty} f(b_n)$$

  (or one sequence with $f(x_n)$ divergent).

「存在」要對**所有**數列成立；「不存在」只需要找到反例數列。

Example 3.7: $f(x) = \sin \frac{1}{x}$ Has No Limit at 0



Choose

$$a_n = \frac{2}{(4n+1)\pi}, \qquad b_n = \frac{2}{(4n+3)\pi}.$$

Then $a_n, b_n \to 0$, but

$$f(a_n) = 1 \to 1, \qquad f(b_n) = -1 \to -1.$$

By Theorem 3.6, $\lim_{x\to 0} f(x)$ does not exist.

選數列的技巧：讓 $\frac{1}{x}$ 落在 $\sin$ 的最大值 $\frac{\pi}{2} + 2n\pi$ 與最小值 $\frac{3\pi}{2} + 2n\pi$。

Limit Theorems

以下都是由 Theorem 3.6 加上數列的定理得到

Theorem 3.8 (Algebra of limits)

If $f(x) \to L$ and $g(x) \to M$ as $x \to a$, then

$$f+g \to L+M, \qquad \alpha f \to \alpha L, \qquad fg \to LM, \qquad \frac{f}{g} \to \frac{L}{M} \ (M \neq 0).$$

Theorem 3.9 (Squeeze Theorem for Functions)

i) If $g(x) \leq h(x) \leq f(x)$ on $I \setminus \{a\}$ and $f, g \to L$, then $h(x) \to L$.

ii) If $|g(x)| \leq M$ on $I \setminus \{a\}$ and $f(x) \to 0$, then $f(x)g(x) \to 0$.

Theorem 3.10 (Comparison Theorem for Functions)

If both limits exist and $f(x) \leq g(x)$ on $I \setminus \{a\}$, then $\lim\limits_{x \to a} f(x) \leq \lim\limits_{x \to a} g(x)$.

Using the Limit Theorems

Example 3.11

$$\lim_{x\to 1}\frac{x-1}{3x+1}=\frac{\lim_{x\to 1}(x-1)}{\lim_{x\to 1}(3x+1)}=\frac{0}{4}=0$$

No ε–δ needed: $x-1\to 0$ and $3x+1\to 4\neq 0$.

"Bounded $\times$ zero" (Theorem 3.9 ii)

$$\lim_{x\to 0} x\sin\frac{1}{x}=0, \qquad \text{since } \left|\sin\frac{1}{x}\right|\le 1 \text{ and } x\to 0.$$

Which tool?

- The problem says "using Definition 3.1" $\Rightarrow$ ε–δ.
- Otherwise: simplify (Remark 3.4) $\rightarrow$ limit theorems $\rightarrow$ squeeze.
- Limit does not exist $\Rightarrow$ two sequences (Theorem 3.6).

注意 $\sin\frac{1}{x}$ 本身沒有極限，但乘上趨近 0 的 x 之後就有極限。

§ 3.1 Common Mistakes 常見錯誤

Mistake 1: δ depends on x

"Take $\delta = \varepsilon/|x+2|$." δ may depend on ε and a only. Bound $|x+2|$ first, e.g. $\delta = \min\{1, \varepsilon/4\}$.

Mistake 2: checking only one sequence

Theorem 3.6 needs $f(x_n) \to L$ for **every** $x_n \to a$ ($x_n \neq a$).

Mistake 3: quotient rule with $M = 0$

$\lim \frac{f}{g} = \frac{\lim f}{\lim g}$ is only valid when $\lim g \neq 0$. Simplify first (Remark 3.4).

Mistake 4: strict inequalities in the limit

$f(x) < g(x)$ only gives $\lim f \leq \lim g$. E.g. $0 < x^2$ for $x \neq 0$, but both limits at 0 are 0.

§ 3.2

One-Sided Limits and Limits at Infinity

單邊極限與無窮遠處的極限

One-Sided Limits

Motivation: $f(x) = \sqrt{x-1}$ is only defined for $x \geq 1$, so Definition 3.1 does not apply at $a = 1$ (no open interval around 1).

Definition 3.12

i) **Right-hand limit:** f is defined on an open interval I with left endpoint a, and $\forall \varepsilon > 0\ \exists \delta > 0$ such that

$$a < x < a + \delta \implies |f(x) - L| < \varepsilon.$$

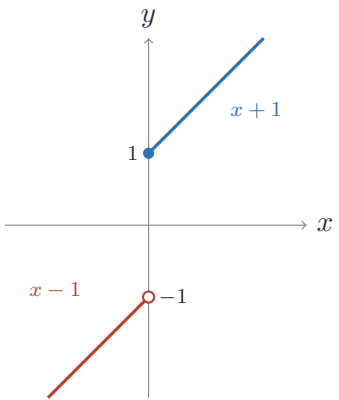
Notation: $f(a+) := L = \lim_{x \to a^+} f(x)$.

ii) **Left-hand limit:** similarly with $a - \delta < x < a$; notation $f(a-) = \lim_{x \to a^-} f(x)$.

- All limit theorems of § 3.1 (3.8, 3.9, 3.10) still hold for one-sided limits.

右極限只看 a 的右邊、左極限只看左邊。

One-Sided vs. Two-Sided



Example 3.13 (i)

$f(x) = x + 1$ for $x \geq 0$, $f(x) = x - 1$ for $x < 0$:

$$f(0+) = 1, \qquad f(0-) = -1.$$

Hence $\lim_{x \to 0} f(x)$ does not exist.

Theorem 3.14

$$\lim_{x \to a} f(x) = L \iff \lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x) = L.$$

雙邊極限存在 $\iff$ 左右極限都存在**而且相等**；分段函數的接點常用此法判斷。

Limits at Infinity and Infinite Limits

Definition 3.15

i) $f(x) \to L$ as $x \to \infty$: $(c, \infty) \subset \mathrm{Dom}(f)$ for some $c > 0$, and $\forall \varepsilon > 0\ \exists M \in \mathbb{R}$ such that $x > M \Rightarrow |f(x) - L| < \varepsilon$.

ii) $f(x) \to \infty$ as $x \to a$: $I \setminus \{a\} \subset \mathrm{Dom}(f)$ for an open $I \ni a$, and $\forall M \in \mathbb{R}\ \exists \delta > 0$ such that $0 < |x - a| < \delta \Rightarrow f(x) > M$.

$x \to a$	$0 < \lvert x - a \rvert < \delta$	$f(x) \to L$	$\lvert f(x) - L \rvert < \varepsilon$
$x \to a^+$	$a < x < a + \delta$	$f(x) \to \infty$	$f(x) > M$
$x \to a^-$	$a - \delta < x < a$	$f(x) \to -\infty$	$f(x) < M$
$x \to \infty$	$x > M$		
$x \to -\infty$	$x < M$		

任選一種「x 的趨近方式」配上一種「$f(x)$ 的目標」，就組合出一個定義。

Examples 3.16: Proof Techniques

(i) $\frac{1}{x} \to 0$ as $x \to \infty$

Given $\varepsilon > 0$, set $M = \frac{1}{\varepsilon}$. Then $x > M \Rightarrow \left|\frac{1}{x}\right| < \frac{1}{M} = \varepsilon$.

(ii) $\lim_{x \to 1^-} \dfrac{x+2}{2x^2 - 3x + 1} = -\infty$: strategy

1. WLOG $M < 0$ (we only need it for very negative M).
2. **Sign of the denominator:** $2x^2 - 3x + 1 = (2x-1)(x-1)$ is negative and $\to 0$ as $x \to 1^-$.
3. **Bound the numerator:** for $0 < x < 1$, $2 < x + 2 < 3$.
4. Choose δ so that the denominator lies in $\left(\frac{2}{M}, 0\right)$; then $f(x) < M$.

處理 $\pm\infty$ 的極限：先決定分母的正負號，再把分子夾在兩個正常數之間。

Unified Sequential Characterization

Notation (4): let $a \in [-\infty, \infty]$ and I an open interval that contains a or has a as an endpoint. Then $\lim\limits_{\substack{x \to a \\ x \in I}} f(x)$ means

$$\lim_{x \to a}, \qquad \lim_{x \to a^+}, \qquad \lim_{x \to a^-}, \qquad \lim_{x \to \pm\infty}$$

depending on where a sits relative to I.

Theorem 3.17

$$\lim_{\substack{x \to a \\ x \in I}} f(x) = L \iff f(x_n) \to L \text{ for all } x_n \in I \text{ with } x_n \neq a,\ x_n \to a.$$

Here a and L may be extended real numbers.

- Consequence: function analogues of Theorem 2.15 / Corollary 2.16 (limits involving $\pm\infty$).

Theorem 3.6 推廣到單邊、無窮遠、無窮大的所有情形：九種情況一次搞定。

Example 3.18: Rational Functions at ∞

Divide by the highest power

$$\lim_{x\to\infty} \frac{2x^2-1}{1-x^2} = \lim_{x\to\infty} \frac{2-\frac{1}{x^2}}{-1+\frac{1}{x^2}} = \frac{2-0}{-1+0} = -2.$$

Why it works

- $\frac{1}{x} \to 0$ (Example 3.16 i) and the product rule give $\frac{1}{x^m} \to 0$ for every $m \in \mathbb{N}$.
- After dividing, the denominator tends to a **nonzero** limit, so Theorem 3.8 applies.

Rule of thumb for $\frac{P(x)}{Q(x)}$ as $x \to \infty$

$\deg P < \deg Q$: limit 0; $\deg P = \deg Q$: ratio of leading coefficients;
$\deg P > \deg Q$: $\pm\infty$.

分子分母同除以「分母的最高次」；$x \to -\infty$ 時要特別注意正負號。

§ 3.2 Common Mistakes 常見錯誤

Mistake 1: one-sided limits exist ⇒ limit exists

Both must exist **and be equal** (Theorem 3.14). Example 3.13: $f(0+) = 1 \neq -1 = f(0-)$.

Mistake 2: treating ∞ as a number

$\infty - \infty = 0$, $\quad \dfrac{\infty}{\infty} = 1$, $\quad 0 \cdot \infty = 0$. These are indeterminate; simplify first.

Mistake 3: ignoring the sign near a vertical asymptote

$\dfrac{1}{x-1} \to +\infty$ as $x \to 1^+$ but $\to -\infty$ as $x \to 1^-$; so $\lim_{x\to 1} \frac{1}{x-1}$ does not exist (even as an extended real number).

Mistake 4: writing $|f(x) - \infty| < \varepsilon$

For $f(x) \to \infty$ the condition is $f(x) > M$, not an ε-estimate.

Think About It

動動腦

$$\lim_{x \to \infty} \left(\sqrt{x^2 + x} - x \right) = ?$$

Think About It: Answer

Structure: "$\infty - \infty$" form $\Rightarrow$ rationalize, then divide by the highest power.

$$\begin{aligned}
\sqrt{x^2+x} - x &= \frac{(\sqrt{x^2+x} - x)(\sqrt{x^2+x} + x)}{\sqrt{x^2+x} + x} && \text{multiply by the conjugate} \\
&= \frac{x}{\sqrt{x^2+x} + x} && (x^2+x) - x^2 = x \\
&= \frac{1}{\sqrt{1+\frac{1}{x}} + 1} && \text{divide by } x > 0
\end{aligned}$$

Since $\frac{1}{x} \to 0$ and $\sqrt{\cdot}$ is continuous at 1, Theorem 3.8 gives

Answer

$$\lim_{x\to\infty} \left(\sqrt{x^2+x} - x\right) = \frac{1}{1+1} = \frac{1}{2}.$$

除以 x 時用到 $x > 0$，所以 $\sqrt{x^2} = x$；若 $x \to -\infty$ 就要改成 $\sqrt{x^2} = -x$。

One-Page Summary

Def 3.1	$\forall \varepsilon > 0\ \exists \delta > 0 :\ 0 < \lvert x - a \rvert < \delta \Rightarrow \lvert f(x) - L \rvert < \varepsilon$
Finding δ	factor out $\lvert x - a \rvert$, restrict $\delta \leq 1$, $\delta = \min\{1, \varepsilon / C\}$
Thm 3.6 / 3.17	$\lim f(x) = L \iff f(x_n) \to L$ for every $x_n \to a$, $x_n \neq a$
No limit	two sequences with different limits of $f(x_n)$
Thm 3.8–3.10	algebra of limits, squeeze, comparison
Thm 3.14	$\lim_{x \to a} f = L \iff f(a+) = f(a-) = L$
Def 3.15	$x > M$ for $x \to \infty$; $f(x) > M$ for $f \to \infty$

Watch out: δ must not depend on x one sequence $\not\Rightarrow$ limit exists ∞ is not a number